Intermediate Microeconomics

Week 3 · Day 4 — Externalities, Public Goods, and the Commons

Summer School

When even a competitive market gets it wrong: costs and benefits that spill onto others.

A different kind of failure

All week, markets failed because of *market power* — a firm restricting output to raise price. Today the market fails for a new reason: **prices do not capture all the costs and benefits**.

The theme

A choice can impose costs on — or provide benefits to — people who are not party to the transaction. Left to itself, the market over- or under-provides.

- **Negative externality**: a factory's pollution harms neighbours (too much produced).
- **Public good**: a lighthouse benefits every ship (too little provided — free riding).
- **The commons**: a shared fishery gets over-harvested (too much extracted).

Learning goals — Day 4

1. Distinguish **private** from **social** marginal cost, and locate the *efficient* quantity under a negative externality.
2. Measure the **deadweight loss** of an externality and design a **Pigouvian tax** to correct it.
3. Explain the **free-rider problem** and classify goods by rivalry and excludability.
4. Find the efficient level of a **public good** (the Samuelson condition, $\sum MB = MC$) and show why private provision *under*-provides.
5. Analyse the **tragedy of the commons**: why open access leads to over-use.

Private vs. social cost

A competitive steel market has (inverse) demand and *private* marginal cost

$$P = 100 - Q \quad \text{(marginal benefit)}, \qquad MPC = 10 + Q.$$

Each tonne also dumps pollution worth \$20 of harm on others — a **marginal external cost** $MEC = 20$, so $MSC = MPC + MEC = 30 + Q$.

The market ignores the harm

Firms equate demand to *private* cost: $100 - Q = 10 + Q \Rightarrow \boxed{Q_{mkt} = 45}$, $P = 55$.

Society counts the harm

The efficient quantity equates demand to *social* cost: $100 - Q = 30 + Q \Rightarrow \boxed{Q^* = 35}$, $P = 65$.

The market makes too much

$45 > 35$: each unit between 35 and 45 costs society more (MSC) than anyone values it (MB).

The deadweight loss of over-production

Over the range $Q \in [35, 45]$ the social cost exceeds the marginal benefit; the wasted surplus is a triangle:

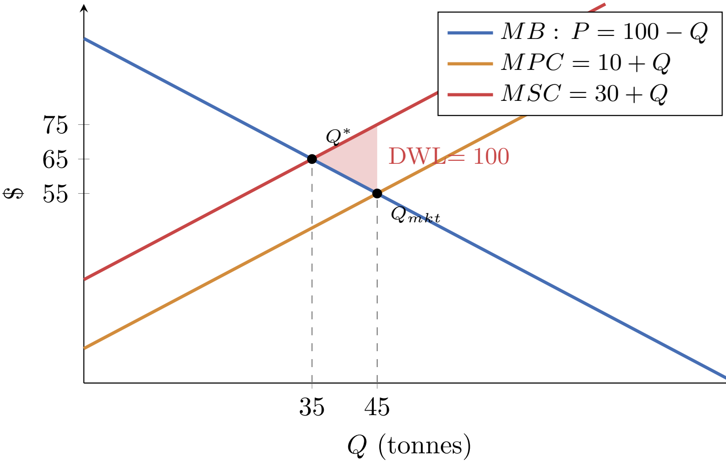
Deadweight loss

$$DWL = \frac{1}{2} \underbrace{(45 - 35)}_{\text{extra units}} \times \underbrace{20}_{MEC} = 100.$$

Reading the picture (next slide)

- At $Q^* = 35$: $MSC = MB$, no waste.
- At $Q_{mkt} = 45$: $MSC = 75 > MB = 55$, a \$20 gap.
- The triangle is the sum of these gaps.

The deadweight loss — the picture



The market produces $Q_{mkt} = 45$; efficiency wants $Q^* = 35$. The shaded triangle is the wasted surplus.

The Pigouvian tax — make the polluter pay

Set a per-unit tax equal to the marginal external cost: $t = MEC = 20$.

The tax internalises the externality

The firm's *private* cost becomes $MPC + t = 30 + Q$, which is exactly MSC. The market now equates

$$100 - Q = 30 + Q \;\Rightarrow\; Q = 35 = Q^*.$$

The deadweight loss of 100 disappears, and the tax raises revenue $20 \times 35 = 700$.

Other remedies

- **Quotas / standards**: cap output at 35 directly.
- **Tradable permits**: issue 35 rights and let the market price the harm.
- **Coase**: with clear property rights and cheap bargaining, the parties can reach Q^* *without* government — but only with few parties and low transaction costs.

Four kinds of goods

Two properties decide how a good behaves in a market:

- **Rival**: my use reduces yours (an apple) vs. *non-rival* (a radio broadcast).
- **Excludable**: I can stop non-payers (a cinema) vs. *non-excludable* (street lighting).

	Excludable	**Non-excludable**
Rival	Private good (food, clothes)	Common resource (fish, pasture)
Non-rival	Club good (cinema, cable)	Public good (defence, lighthouse)

Today's trouble

Markets handle the top-left well. The two shaded cells fail: public goods (under-provided) and common resources (over-used).

The free-rider problem

A **public good** is non-rival and non-excludable: once provided, everyone enjoys it and no one can be shut out.

The incentive to free-ride

Because I benefit *whether or not I pay*, my best move is to let *others* pay and enjoy the good for free. But everyone reasons the same way — so **too little** (or nothing) gets provided.

Familiar logic, familiar examples

- Like a Prisoner's Dilemma: contributing is costly, free-riding is dominant, yet all prefer the good provided.
- Examples: national defence, clean air, a public radio station, open-source software, herd immunity.

How much of a public good *should* be provided, and how much does voluntary contribution deliver?

Efficient provision: the Samuelson condition

A town of two residents, A and B, chooses the size Q of a public good (say park hectares):

$$MB_A = 40 - Q, \qquad MB_B = 60 - Q, \qquad MC = 30.$$

Add benefits *vertically*

Because the good is non-rival, *everyone* consumes the same Q at once, so society's marginal benefit is the **sum** of individual marginal benefits (not the horizontal sum used for private goods):

$$\sum MB = MB_A + MB_B = (40 - Q) + (60 - Q) = 100 - 2Q.$$

Efficient level: $\sum MB = MC$

$$100 - 2Q = 30 \implies \boxed{Q^* = 35}.$$

Society should provide 35 — even though *neither* resident alone values the 35th unit at its $30 cost.

Why private provision falls short — the argument

Left to voluntary contributions, each resident provides only up to where *their own* marginal benefit meets the cost:

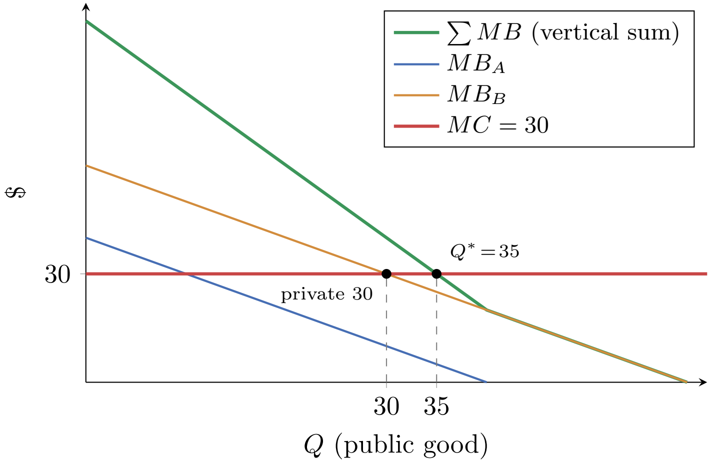
- Resident A: $40 - Q = 30 \Rightarrow Q = 10$.
- Resident B: $60 - Q = 30 \Rightarrow Q = 30$.

B, who values it most, provides 30; A sees 30 already there (where $MB_A = 40 - 30 = 10 < 30$) and **contributes nothing** — A free-rides.

Under-provision

Private outcome $Q = 30 \; < \; Q^* = 35$. The market stops where *one* person's benefit equals cost, ignoring that *everyone* benefits at once.

Why private provision falls short — the picture



Efficiency sums benefits *vertically* ($Q^* = 35$); voluntary provision stops at one person's benefit (30).

The tragedy of the commons — the numbers

A common pasture is open to all villagers. Each cow yields a value that *falls* as the total herd N grows (the field gets crowded):

$$\text{value per cow} = 100 - N, \qquad \text{cost per cow} = 20.$$

The efficient herd (a single owner)

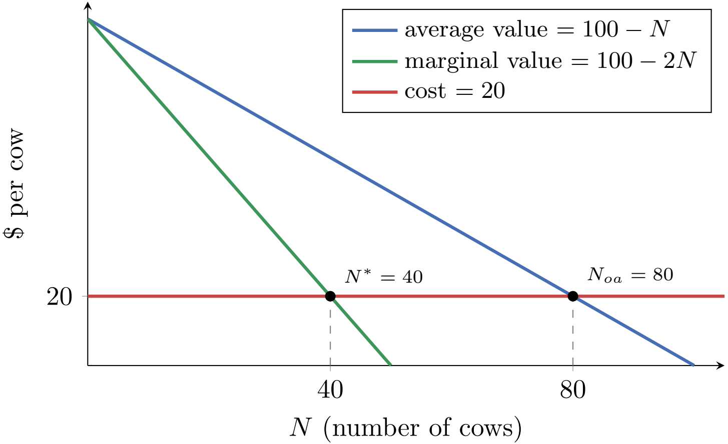
Maximise total village profit $N(100 - N) - 20N = 80N - N^2$; FOC $100 - 2N = 20 \Rightarrow \boxed{N^* = 40}$ (marginal value = cost).

Open access (each villager decides)

A villager adds a cow whenever its *own* value covers its cost: $100 - N = 20 \Rightarrow \boxed{N_{oa} = 80}$ (average value = cost). The herd **doubles**.

Each villager ignores that their extra cow lowers the value of *every other* cow — a negative externality on fellow users.

The tragedy of the commons — the picture



Efficiency sets *marginal* value = cost ($N^* = 40$); open access drives entry until *average* value = cost ($N_{oa} = 80$).

Remedies: property rights (enclosure), quotas / catch limits, use taxes, or community rules (Ostrom).

Externalities and public goods: one idea, three faces

	What spills over	**Market's mistake / fix**
Negative externality	a *cost* onto others	over-produces; tax at $t = MEC$
Public good	a *benefit* to all, non-excludable	under-provides (free-riding); public provision
Common resource	a cost onto other *users*	over-used; property rights / quotas

In every case the price a decision-maker faces omits a cost or benefit borne by others. Efficiency requires *internalising* it — through taxes, rights, or collective provision.

Day 4 — and week — checkpoint

Today you can

- find the efficient quantity under a negative externality and its deadweight loss;
- design a Pigouvian tax $t = MEC$ to restore efficiency;
- apply the Samuelson condition $\sum MB = MC$ and explain free-riding;
- analyse the tragedy of the commons ($N_{oa} > N^*$).

The week in one sentence

Markets can fail through **market power** (Days 1–3: duopoly, strategy, collusion) *and* through **externalities** (Day 4) — and in each case the tools of micro tell us both *why* and *by how much*, and point to a remedy.

In-class exercises — Day 4

Time for the in-class exercises.

- **Exercise 1** — a negative externality: efficient quantity, deadweight loss, and the corrective tax.
- **Exercise 2** — a public good (Samuelson condition and free-riding) and a short tragedy-of-the-commons calculation.

Work in pairs; be ready to state the efficient quantity and the size of the market's error in each case.